

Homework 3 (Due Friday, Week 6):

1. Find the vertex and axis of symmetry of $f(x) = -x^2 + 6x - 5$.
2. Solve $3x^2 - 12x + 9 = 0$ using the quadratic formula.
3. Sketch the graph of $f(x) = x^2 - 2x - 8$. Label the vertex and intercepts.
4. A ball is thrown upward with velocity modeled by $h(t) = -16t^2 + 64t + 80$.
 - Find the time when the ball reaches maximum height.
 - What is the maximum height?